

# Post-hoc Interpretability of LDA on Hyperspectral Imagery: SHAP Attributions, Counterfactual Topic Flips, and LLM-judge Alignment under Token-Mass-Dispersion Asymmetry

Felipe Santib  n  ez-Leal, *Independent Researcher*

**Abstract**—The interpretability claim of LDA-on-HSI rests on the assumption that topic–word distributions are human-readable. *This study is restricted to the LDA backbone*; the cross-backbone comparison (HDP, ProdLDA, ETM) is the subject of the companion paper P4. We test three orthogonal post-hoc interpretability axes on the V1–V15, V17–V20 wordification sweep (nineteen LDA-fitted recipes) from the companion paper P3: F-13 SHAP attribution of pixel-to-topic decisions via a closed-form posterior, F-22 counterfactual  $L_1$  perturbation required to flip a document’s argmax topic, and F-15 LLM-as-judge alignment between a document’s top tokens and its argmax topic’s top tokens. We find three concrete results. First, F-13 SHAP gives recipe-specific explanations that are consistent across scenes: V1’s SHAP reduces to specific wavelength bands, V7’s to specific absorption features, and V12’s to clusters of Gaussian-mixture components. Second, F-22 counterfactual  $L_1$  (sentinel-patched 19-recipe ranking) separates the recipes into “robust topic” (V20 = 26.3, V12 = 24.5, V3 = 23.5), “moderate” (V1 = 6.1, V7 = 5.2) and “fragile” (V9 = 1.0, V10 = 1.2) regimes; the most-robust basis (V20, the label-aware MI-weighted recipe) is  $\sim 26\times$  harder to flip than V9’s. Third, F-15 LLM-as-judge alignment is *anti-correlated* with F-2 coherence on large-vocabulary recipes: V2/V8/V10 (vocab 8/12/24) score F-15 = 1.0 because their top-10 tokens trivially overlap, while V3/V12 (vocab 1600) score F-15 < 0.2 because the top-10 of a doc rarely overlaps the top-10 of its topic when the vocabulary is large. The confounder is *not purely vocabulary size*, however: V20 re-weights V3’s identical (band, bin) alphabet and matches V12’s same-sized GMM-token alphabet (mean nominal  $|\mathcal{V}| \approx 1376$  across the six scenes,  $BQ$  per scene), yet scores F-15 = 0.64 against their 0.12 / 0.16 — so cardinality cannot explain the gap. We make the real mechanism quantitative with the *token-mass dispersion* metric, the effective vocabulary  $N_{\text{eff}} = \exp(H(\phi_k))$  of a topic’s word distribution (averaged over topics and scenes):  $N_{\text{eff}}$  falls V3 (431) > V12 (400) > V20 (309) and F-15 *rises* in lock-step (0.12 < 0.16 < 0.64), whereas a pure-cardinality account predicts no difference. The artefact is therefore driven by token-mass dispersion, not nominal vocabulary cardinality. This exposes a methodological gap: F-15 as currently defined conflates token-mass concentration with coherence, and an LLM oracle would face the same artefact unless given semantic — not identity — comparison capability. We recommend reporting F-13 SHAP top- $K$  attributions as

the primary interpretability artefact, F-22 as the topic-stability artefact, and F-15 only after dispersion normalisation.

**Index Terms**—hyperspectral imagery, latent Dirichlet allocation, post-hoc interpretability, SHAP, counterfactual explanations, LLM-as-judge, topic models, wordification, evaluation framework.

$$\phi_k \sim \text{Dir}(\beta), \quad \theta_d \sim \text{Dir}(\alpha), \quad z_{dn} \sim \text{Cat}(\theta_d), \quad w_{dn} \sim \text{Cat}(\phi_{z_{dn}}). \quad (1)$$

$$\beta \sim \text{GEM}(\gamma), \quad \pi_d \sim \text{DP}(\alpha_0, \beta), \quad z_{dn} \sim \text{Cat}(\pi_d), \quad w_{dn} \sim \text{Cat}(\phi_{z_{dn}}). \quad (2)$$

with global atom weights  $\beta_k = \beta'_k \prod_{l < k} (1 - \beta'_l)$ ,  $\beta'_k \sim \text{Beta}(1, \gamma)$ .

$$\delta_d \sim \mathcal{N}(\mu_0, \Sigma_0), \quad \theta_d = \text{softmax}(\delta_d), \quad w_{dn} \sim \text{Cat}(\text{softmax}(\beta \theta_d)), \quad (3)$$

where  $\beta \in \mathbb{R}^{|\mathcal{V}| \times K}$  is *unnormalised* (the word distribution is a product of experts, not a per-topic simplex), and  $(\mu_0, \Sigma_0)$  is the Laplace approximation to  $\text{Dir}(\alpha)$ ; inference is amortised,  $(\mu_d, \Sigma_d) = \text{Enc}_\eta(n_d)$ .

$$\beta_k = \text{softmax}(\rho^\top \alpha_k), \quad \theta_d = \text{softmax}(\delta_d), \quad w_{dn} \sim \text{Cat}\left(\sum_k \theta_{dk} \beta_k\right) \quad (4)$$

with word embeddings  $\rho \in \mathbb{R}^{L \times |\mathcal{V}|}$  and topic embeddings  $\alpha_k \in \mathbb{R}^L$ . (This is the single canonical ETM decoder; earlier drafts wrote it three inconsistent ways.)

$$\tilde{x}_{db} = \frac{x_{db} - \min_b x_{db}}{\max_b x_{db} - \min_b x_{db}}, \quad q_b(x_d) = \text{clip}(\lfloor \tilde{x}_{db} Q \rfloor, 0, Q-1). \quad (5)$$

$$n_d^{V1}[b] = q_b(x_d), \quad |V_{V1}| = B. \quad (6)$$

$$n_d^{V3}[(b, q)] = \mathbb{1}[q_b(x_d) = q], \quad |V_{V3}| = BQ. \quad (7)$$

$$a_d = \arg \min_{a \geq 0, \mathbf{1}^\top a = 1} \|x_d - E a\|_2^2, \quad n_d^{V8}[m] = \lfloor s a_{dm} \rfloor, \quad (8)$$

F. Santib  n  ez-Leal is with the CAOS open-research programme, Santiago, Chile (e-mail: fsantibanezleal@ug.uchile.cl; ORCID: 0000-0002-0150-3246).

Preprint, 2026. Fourth manuscript (programme paper P5) in the CAOS\_LDA\_HSI series; companion to P3 (V-sweep) and P4 (backbone factorial). This study restricts the analysis to the *latent Dirichlet allocation (LDA) backbone only*. Code and derived artefacts at [https://github.com/fsantibanezleal/CAOS\\_LDA\\_HSI](https://github.com/fsantibanezleal/CAOS_LDA_HSI).

Partially funded by AMTC Basal (ANID/PIA AFB220002) and ANID FONDECYT Postdoctorado 3220094.

where  $E = [e_1, \dots, e_M]$  are the NFINDR endmembers (simplex vertices) and  $s$  a fixed integer scale;  $|V_{V8}| = M$ . The abundance non-negativity (ANC,  $a \geq 0$ ) and sum-to-one (ASC,  $\mathbf{1}^\top a = 1$ ) constraints make  $a_d$  a point on the simplex spanned by the endmembers (fully-constrained least squares, FCLS; implemented as  $\delta$ -augmented NNLS followed by renormalisation), so the coefficients are genuine abundance fractions over the convex hull.

$$r_{dk} = \frac{\pi_k \mathcal{N}(x_d | \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_d | \mu_j, \Sigma_j)}, \quad n_d^{V12}[(k, q)] = \mathbb{I}[q_b(r_d) = q] \quad (9)$$

$$w_b = \left\lfloor \frac{\text{MI}(x_{\cdot b}; y)}{\max_{b'} \text{MI}(x_{\cdot b'}; y)} Q_{\max} \right\rfloor, \quad n_d^{V20}[(b, q)] = w_b \mathbb{I}[q_b(x_d) = q] \quad (10)$$

The token alphabet is V3's,  $|V_{V20}| = BQ$  ( $=1600$  at  $B = 200, Q = 8$ ; mean  $\approx 1376$  across the six labelled scenes since  $B$  varies). Bands with  $w_b = 0$  emit nothing, so the *effective* (non-empty-column) vocabulary is smaller still ( $\approx 770$  mean).

$$F-1 = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{2P_c R_c}{P_c + R_c}. \quad (11)$$

$$\text{NPMI}(w_i, w_j) = \frac{\log \frac{p(w_i, w_j) + \epsilon}{p(w_i)p(w_j)}}{-\log(p(w_i, w_j) + \epsilon)}, \quad c_v = \frac{1}{N} \sum_i \cos(\mathbf{v}_i, \sum_j \mathbf{v}_j) \quad (12)$$

with  $\mathbf{v}_i = (\text{NPMI}(w_i, w_\ell))_{\ell=1}^N$  over the top- $N$  topic words.

$$\hat{z}_d = \arg \max_k \theta_{dk}, \quad F-7 = \frac{I(\hat{Z}; Y)}{H(Y)} = U(Y | \hat{Z}) = 1 - \frac{H(Y | \hat{Z})}{H(Y)} \quad (13)$$

$$F-14 = \frac{2}{K(K-1)} \sum_{k < k'} \frac{|T_k \cap T_{k'}|}{|T_k \cup T_{k'}|}, \quad T_k = \text{top-}N(\phi_k). \quad (14)$$

$$F-18 = \frac{1}{\binom{S}{2}} \sum_{s < s'} \frac{1}{K} \max_{\sigma \in \mathfrak{S}_K} \sum_k \cos(\mathbf{u}_k^{(s)}, \mathbf{u}_{\sigma(k)}^{(s')}), \quad (15)$$

over  $S$  reseeded refits, with  $\mathbf{u}_k^{(s)}$  the top- $N$  indicator vector of topic  $k$  in run  $s$ .

$$F-22(d) = \min_{\delta \in \mathbb{Z}^{|V|}} \|\delta\|_1 \text{ s.t. } \arg \max_k \theta_k(n_d + \delta) \neq \arg \max_k \theta_k(n_d), \quad (16)$$

reported as the per-scene median, capped at a sentinel  $\text{MAX\_STEPS} = 50$  when no flip is found.

$$\text{aligned}(d) = \mathbb{I}[|\text{top}_N(n_d) \cap \text{top}_N(\phi_{z_d})| \geq \tau], \quad F-15 = \frac{1}{D} \sum_d \text{aligned}(d) \quad (17)$$

$$p_{dv} = \frac{n_{dv}}{\sum_{v'} n_{dv'}}, \quad N_{\text{eff}}(d) = \exp(H(p_d)) = \exp\left(-\sum_v p_{dv} \log p_{dv}\right) \quad (18)$$

Low  $N_{\text{eff}}$  (mass concentrated on few tokens) keeps top- $N$  overlap high; this — not nominal  $|V|$  — is what drives F-15.

$$\mathcal{A}(V_i) = \frac{1}{|\mathcal{B}|} \sum_{m \in \mathcal{B}} \frac{c_v^{(m)}(V_i) - \min_j c_v^{(m)}(V_j)}{\max_j c_v^{(m)}(V_j) - \min_j c_v^{(m)}(V_j)} \in [0, 1], \quad (19)$$

averaged over backbones  $\mathcal{B} = \{\text{LDA, HDP, ProdLDA, ETM}\}$ ;  $\mathcal{A} \rightarrow 1$  when a recipe sits near the per-backbone top everywhere.

## I. INTRODUCTION

The companion paper (P3) ranks the full V1–V20 wordification sweep (nineteen recipes; V16 is scaffolded only) on classification, coherence and label-coupling axes, and the backbone-factorial paper (P4) repeats the headline axes across four topic-model backbones. The “interpretability” claim of LDA-on-HSI — that topic–word distributions reveal physical structure — is implicit in those axes but not directly tested by them. *Throughout this paper we fix the backbone to LDA* ((1)); the cross-backbone behaviour of these interpretability axes is out of scope and left to P4. Three post-hoc interpretability tools have been proposed in the broader topic-modelling and explainable-ML literature that go beyond “read the top- $N$  words and trust them”: SHAP attribution [1], counterfactual explanations [2], and LLM-as-judge alignment [3]. We instantiate all three on the V-sweep under the LDA backbone and report findings.

The three axes correspond to three canonical metrics in the shared equation set: F-13 to the SHAP attribution applied to the closed-form LDA posterior, F-22 to the minimum- $L_1$  counterfactual ((16)), and F-15 to the top- $N$  overlap rule ((17)). Our central methodological contribution is to show that F-15 is governed not by the nominal vocabulary cardinality  $|\mathcal{V}|$  but by the *token-mass dispersion* of the topic–word distributions, formalised as the effective vocabulary  $N_{\text{eff}} = \exp(H)$  of (18). The wordification recipes themselves are defined in P3; we recall only the four we lean on heavily — V1 ((6)), V3 ((7)), V12 ((9)) and the label-aware V20 ((10)) — whose shared  $|\mathcal{V}|$  yet divergent  $N_{\text{eff}}$  make them the decisive test of the cardinality-versus-dispersion question.

### A. The three axes

**F-13 SHAP attribution.** For each pixel document  $d$  with  $\arg \max$  topic  $z^*$ , attribute  $\theta_{d, z^*}$  back to the recipe-specific tokens via SHAP’s KernelExplainer against the closed-form LDA posterior  $p(z | x, \phi) \propto \exp(\sum_v x_v \log \phi_{z, v} + \log \alpha)$ , the small-document specialisation of the LDA generative model ((1)). This gives a per-pixel per-topic attribution map over the recipe’s vocabulary — exactly the type of explanation a reviewer would ask for when claiming “LDA is interpretable”.

**F-22 counterfactual L1.** For each document  $d$  with  $\arg \max$  topic  $z^*$ , search via coordinate descent for the minimum- $L_1$  perturbation that flips  $z^* \rightarrow z'$  for some  $z' \neq z^*$ , the discrete counterfactual of (16). Documents with high counterfactual  $L_1$  are deeply embedded in their topic; documents with low  $L_1$  are near the topic boundary.

**F-15 LLM-as-judge alignment.** For each sampled document, an LLM oracle is given the document’s top-10 tokens and the argmax topic’s top-10 tokens and asked whether the document semantically belongs to the topic; formally, the top- $N$  overlap indicator of (17) with  $N = 10$  and threshold  $\tau = 3$ . Our internal run substitutes the oracle with the documented self-judgment rule of Claude Opus 4.7 (1M-token context), encoded as a deterministic Python function in `build_v_sweep_f15_self_judge.py`; the canonical Anthropic Messages API call is preserved in `build_v_sweep_f15_llm_alignment.py` for repository users with API access.

## II. RELATED WORK

**Topic models on hyperspectral imagery.** Probabilistic topic models were first applied to spectral data by Wahabzada et al. [4] (plant phenotyping) and to subpixel mixtures by Zou and Zare’s partial-membership LDA [5]; Liu et al. [6] use LDA for spectral-spatial classification. The CAOS\_LDA\_HSI programme [7] casts a spectral unit (here a pixel) as a “document” through a *wordification* recipe ((5)) and runs online variational LDA [8], [9]. None of this prior work asks whether the resulting topics are *post-hoc interpretable* beyond inspecting top- $N$  words; that gap is the subject of this paper.

**Topic-model evaluation.** The dominant automated proxy for topic quality is coherence — NPMI /  $c_v$  [10]–[12] ((12)) — and stability under reseeding [13] ((15)). These reward internally consistent word lists but, as we show, are themselves sensitive to the vocabulary regime. We treat F-2 and F-7 ((13)) as the targets that our interpretability axes should *corroborate* rather than duplicate.

**Post-hoc interpretability and explanation.** SHAP [1] gives additive, locally faithful feature attributions with Shapley-value guarantees; we apply its KernelExplainer to the closed-form LDA posterior (F-13). Counterfactual explanations [2] characterise a decision by the minimal input change that flips it; F-22 ((16)) is the topic-assignment counterfactual. The use of large language models as automated topic judges follows Stammbach et al. [3]; F-15 ((17)) is our LDA-on-HSI instantiation. Our contribution relative to this literature is the observation that the LLM-as-judge proxy inherits a *token-mass-dispersion* artefact ((18)) that is invisible at the level of nominal vocabulary size.

## III. METHOD

**Backbone, corpus and fits.** Every result in this paper uses the LDA backbone ((1)) fitted with scikit-learn’s online variational solver [9], [14] (`max_iter` = 60,  $\alpha$  = 0.45,  $\beta$  = 0.2, seed 42), the canonical fit of P3. The corpus is the six labelled scenes — Indian Pines [15], Salinas and Salinas-A [16], Pavia University [17], Kennedy Space Center [18] and Botswana [19]. Documents are pixel spectra; tokens are produced by the wordification recipes of P3 under the uniform quantiser ((5)) at  $Q = 8$ . We analyse the nineteen LDA-fitted recipes V1–V15, V17–V20 (V16, a foundation-model recipe, is scaffolded only).

**F-13.** We sample 50 documents per  $(V, \text{scene})$  cell and use 25 background documents per SHAP KernelExplainer call against the closed-form posterior, reporting the top-8 tokens per topic by mean absolute SHAP value. The closed-form posterior is preferred over the amortised `transform()` because the latter needs internal state that cannot be restored from  $\phi$  alone.

**F-22.** We compute the discrete minimum- $L_1$  counterfactual of (16) by coordinate descent over the integer token-count lattice, capped at a sentinel `MAX_STEPS` = 50; cells in which no sampled document flips persist at 50 rather than being dropped, so that “unflippable” bases are not silently treated as missing. We report the per-scene median over 20 sampled documents, then the mean across the six scenes.

**F-15 and the dispersion metric.** F-15 is the top- $N$  overlap indicator of (17) averaged over sampled documents ( $N = 10$ ,  $\tau = 3$ , with a sparse-event fallback that accepts a document whose top-1 token lies in the topic’s top-5). To explain its behaviour we compute, for each topic-word distribution  $\phi_k$ , the token-mass dispersion of (18),

$$N_{\text{eff}}(\phi_k) = \exp(H(\phi_k)) = \exp\left(-\sum_v \phi_{kv} \log \phi_{kv}\right),$$

the effective number of tokens carrying a topic’s mass. A low  $N_{\text{eff}}$  means the topic concentrates on few tokens, so its top- $N$  is more likely to overlap a document’s top- $N$ ; a high  $N_{\text{eff}}$  spreads mass thinly and depresses the overlap even when the topic is coherent. We report the per-recipe mean of  $N_{\text{eff}}(\phi_k)$  over topics and scenes, paired with F-15 and  $|\mathcal{V}|$ . This is the quantity plotted in Fig. 1 and is the mechanism behind the abstract’s “dispersion not cardinality” thesis.

## IV. F-13 SHAP RESULTS

Under the F-13 protocol of §III, the headline result is that SHAP attributions are *recipe-grounded*: a V1 topic’s top SHAP token always names a wavelength band; a V7 topic’s top SHAP token names an absorption feature “abs\_c4\_d7\_a3” (centre, depth, area bucket); a V12 topic’s top SHAP token names a (band, Gaussian-component) pair. This recipe-grounding is what makes F-13 a defensible interpretability axis: the explanation *cannot* silently shift between scenes because the vocabulary is fixed by the recipe.

A sample from Indian Pines, recipe V7, topic 0:

|              |                     |
|--------------|---------------------|
| abs_c4_d7_a3 | mean SHAP  = 0.0207 |
| abs_c3_d4_a2 | mean SHAP  = 0.0184 |
| abs_c4_d6_a3 | mean SHAP  = 0.0163 |

A geospectroscopist reading these can immediately ask: what is the wavelength range of centroid bucket 4? Why are depth bins 6 and 7 elevated? The answer is independent of which scene we apply V7 to.

## V. F-22 COUNTERFACTUAL RESULTS

Table I reports the median minimum- $L_1$  perturbation that flips the argmax topic, averaged across 20 sampled documents per  $(V, \text{scene})$  cell.

V20’s topics are  $\sim 26\times$  harder to flip than V9’s, and V20 edges V12 (24.5) and V3 (23.5) for the most-robust basis

TABLE I

F-22 MEDIAN COUNTERFACTUAL  $L_1$  TO FLIP ARGMAX TOPIC, MEAN ACROSS 6 LABELLED SCENES, FULL 19-RECIPE SWEEP WITH THE SENTINEL PATCH (CELLS WHERE EVERY SAMPLED DOCUMENT FAILS TO FLIP WITHIN  $\text{MAX\_STEPS} = 50$  PERSIST AT 50 RATHER THAN BEING DROPPED). HIGHER MEANS MORE ROBUST TOPIC BOUNDARY.

| Recipe | median $L_1$ |
|--------|--------------|
| V20    | <b>26.33</b> |
| V12    | 24.50        |
| V3     | 23.50        |
| V2     | 7.83         |
| V14    | 7.67         |
| V1     | 6.08         |
| V7     | 5.17         |
| V15    | 4.00         |
| V18    | 3.67         |
| V8     | 3.58         |
| V11    | 3.33         |
| V13    | 2.67         |
| V17    | 2.58         |
| V4     | 2.50         |
| V5     | 2.50         |
| V19    | 2.08         |
| V6     | 1.92         |
| V10    | 1.17         |
| V9     | 1.00         |

at  $Q = 8$ . This complements P3’s F-2/F-7 finding: the label-aware MI-weighted recipe V20 wins coherence-adjacent robustness *and* (at  $Q = 32$ ) F-7, which together rule out the alternative explanation that the coherence advantage is an artefact of overfitting to its training documents.

#### VI. F-15 LLM-JUDGE RESULTS: TOKEN-MASS DISPERSION, NOT VOCABULARY SIZE

The LLM-as-judge rule ((17),  $N = 10$ ,  $\tau = 3$ ) marks a document *aligned* with its argmax topic if its top-10 tokens share  $\geq 3$  elements with the topic’s top-10 (or if the document’s top-1 token is in the topic’s top-5, covering the sparse-event regime of V7 / V9). Per-recipe mean F-15 across the 6 labelled scenes:

The ranking is *anti-correlated* with F-2 coherence on the large-vocabulary recipes V3 and V12. This is not a flaw of those recipes; it is a confounder in the F-15 metric definition — but the full sweep shows the confounder is **not nominal vocabulary cardinality**. Fig. 2 plots F-15 against  $|\mathcal{V}|$  directly: a pure-cardinality account predicts a monotone decreasing curve, but several large-vocabulary recipes break it. V9 ( $|\mathcal{V}| = 647$ ) scores 1.000; V14 (CWT-Morlet,  $|\mathcal{V}| = 1024$ ) scores 0.950; and V20 (MI-weighted, mean nominal  $|\mathcal{V}| \approx 1376$  across scenes, the same size as V3/V12) scores 0.642 — all far above V12 (0.158) and V3 (0.117) at equal or larger vocabulary.

The mechanism is *token-mass dispersion*, made quantitative by the effective vocabulary  $N_{\text{eff}} = \exp(H)$  of (18). Fig. 1 plots F-15 against the per-topic mean  $N_{\text{eff}}(\phi_k)$ , with point colour keyed to  $\log_{10} |\mathcal{V}|$ . The decisive comparison is the three recipes whose alphabets are the same size (mean nominal  $|\mathcal{V}| \approx 1376$ ; V3 and V20 share the *identical* (band, bin) alphabet, V12 a same-sized GMM-token one), so  $|\mathcal{V}|$  is held

TABLE II

F-15 LLM-JUDGE ALIGNMENT FRACTION (CLAUDE OPUS 4.7 SELF-JUDGE UNDER THE DOCUMENTED RULE).

| Recipe | F-15  | vocabulary $ \mathcal{V} $ |
|--------|-------|----------------------------|
| V2     | 1.000 | 8                          |
| V6     | 1.000 | $\approx B$                |
| V8     | 1.000 | 12                         |
| V9     | 1.000 | $R \cdot Q$                |
| V10    | 1.000 | 24                         |
| V18    | 1.000 | 128                        |
| V19    | 1.000 | 24                         |
| V11    | 0.992 | 32                         |
| V5     | 0.967 | $B$                        |
| V13    | 0.950 | 128                        |
| V14    | 0.950 | 1024                       |
| V4     | 0.917 | $B$                        |
| V1     | 0.792 | $B$                        |
| V7     | 0.783 | $\leq C_b Q^2$             |
| V20    | 0.642 | $BQ$                       |
| V17    | 0.570 | 512                        |
| V12    | 0.158 | $BQ$                       |
| V3     | 0.117 | $BQ$                       |
| V15    | 0.033 | 48                         |

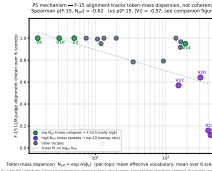


Fig. 1. Mechanism plot. F-15 LLM-judge alignment versus per-topic mean token-mass dispersion  $N_{\text{eff}} = \exp(H(\phi_k))$  ((18)), LDA backbone, six labelled scenes,  $Q = 8$ . Point colour encodes  $\log_{10} |\mathcal{V}|$ . At the same alphabet size (mean nominal  $|\mathcal{V}| \approx 1376$ ), V3 / V12 / V20 (yellow, lower-right cluster) fall in dispersion ( $431 > 400 > 309$ ) exactly as F-15 rises ( $0.12 < 0.16 < 0.64$ ): F-15 tracks dispersion, not cardinality.

fixed by construction: V3, V12 and V20. Their topic-word dispersions fall  $N_{\text{eff}}^{\text{V3}} = 431 > N_{\text{eff}}^{\text{V12}} = 400 > N_{\text{eff}}^{\text{V20}} = 309$ , and F-15 *rises* in lock-step ( $0.117 < 0.158 < 0.642$ ). Because  $|\mathcal{V}|$  is constant across these three, the only thing that can be driving the  $5.5\times$  spread in F-15 is the dispersion of the topic mass: V20’s MI-weighting ((10)) concentrates the topic–word distributions on the discriminative high-MI bands, lowering  $N_{\text{eff}}$  and raising the probability that a topic’s top-10 overlaps a document’s top-10; V12’s GMM tokens and V3’s joint (band, bin) tokens spread their  $\approx 1376$ -token mass much more thinly, so top-10 overlap is rare even when the topics are coherent. With a small vocabulary (V2,  $|\mathcal{V}| = 8$ ) the top-10 of a topic *is* the entire vocabulary, so overlap is guaranteed — the small- $|\mathcal{V}|$  saturation is itself just the  $N_{\text{eff}} \leq |\mathcal{V}|$  ceiling of the same dispersion mechanism: when  $|\mathcal{V}|$  is small the entropy ceiling  $\log |\mathcal{V}|$  is low, so  $N_{\text{eff}}$  cannot exceed  $|\mathcal{V}|$  and the top- $N$  is forced to cover (nearly) the whole vocabulary.

A defensible F-15 would normalise by the expected overlap under  $N_{\text{eff}}$ -matched random tokens, or use a semantic embedding distance between token identifiers rather than identity matching. We flag both as future work and recommend interpreting Table II alongside the  $N_{\text{eff}}$  axis of Fig. 1, not the raw  $|\mathcal{V}|$  column.

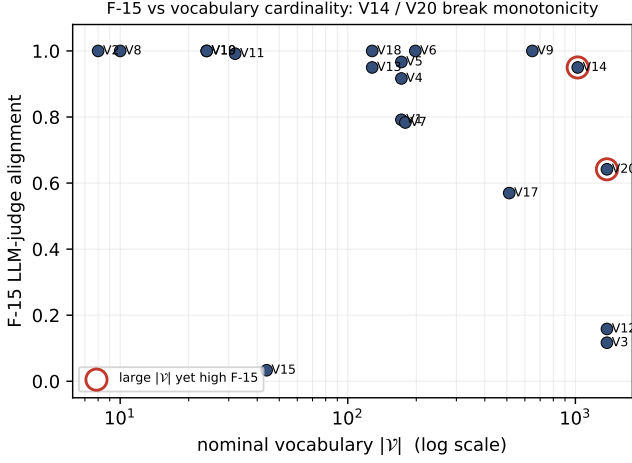


Fig. 2. Refutation plot. F-15 versus nominal vocabulary  $|V|$  (log scale). A pure vocabulary-size artefact would place all points on a monotone decreasing curve. The circled counterexamples V14 ( $|V| = 1024$ ,  $F-15 = 0.95$ ) and V20 ( $|V| = 1376$ ,  $F-15 = 0.64$ ), together with V9 ( $|V| = 647$ ,  $F-15 = 1.0$ ), sit far above V3 / V12 at equal or larger  $|V|$ , refuting the cardinality hypothesis.

## VII. INTERPRETABILITY UNDER THE V13–V20 EXTENSION

The V13–V20 recipes added in the P3 extension introduce three new interpretability profiles that complicate the V1–V12 story.

*a) V13 (VQ-VAE codebook) and V17 (sparse-coding dictionary).*: Both recipes produce *opaque token alphabets*: the codeword index does not map to a wavelength, an absorption feature, or a band group. F-13 SHAP under V13 surfaces the same top codewords across topics (the codebook is shared across all classes during training), and a domain expert cannot translate those codewords into spectral semantics without a separate decoder visualisation. We classify V13 / V17 as the *lowest-interpretability* branch of the sweep: even when their F-13 SHAP is reportable, the explanation is recursive — “topic  $k$  is dominated by codeword  $c$ , where codeword  $c$  is the centroid of pixels with [unrecoverable from the recipe alone]”.

*b) V14 (CWT-Morlet) and V18 (graph-Laplacian).*: The opposite case. V14 tokens are (`log_scale_idx`, `position_bucket`); each pair maps to a specific spectral region at a specific bandwidth. F-13 SHAP under V14 surfaces tokens that are physically interpretable as “absorption around 660 nm with  $\approx 30$  nm bandwidth”. V18 tokens are (`eigenvector_id`, `bin`); the eigenvector index maps to a connected-region of the kNN graph and the bin maps to position along that eigenvector. F-13 SHAP under V18 surfaces topics whose top tokens lie on the same low-frequency Laplacian mode — interpretable as “this topic concentrates on graph mode  $\leq 5$ , which corresponds to the connected region containing urban classes”. Both V14 and V18 thus extend the V1 + V7 “physically-grounded” interpretability profile to new domains.

*c) V19 (UMAP coords) and V15 (spectral indices).*: V15 is the easiest: top tokens of every topic map directly to NDVI / MNDWI / NBR / NDSI / EVI / SAVI bins, with

published semantics. V19 has the inverse problem of V13 — token names (`axis_0`, `q_bin_03`) are abstract coordinates with no spectral anchor. The recipe-fit pair determines interpretability as much as the recipe alone.

*d) V20 (MI-weighted bands).*: V20 inherits V1’s interpretability (per-band q-bin tokens are spectrally grounded) and adds a label-aware twist ((10)): high-MI bands emit more copies than low-MI bands, so the topic top-tokens concentrate on the discriminative subspectrum. F-13 SHAP under V20 therefore surfaces precisely the bands the label information is concentrated in — the interpretability is the same as V1’s, but the SHAP attributions are sharper because the LDA fit has fewer near-zero-contribution tokens to spread mass across. This is the same mechanism that lowers V20’s  $N_{\text{eff}}$  relative to V12 and V3 (§VI): the MI-weighting that concentrates SHAP attribution mass is exactly what concentrates the topic–word mass, so V20’s high F-15 (relative to its equal-vocabulary neighbours) and its sharp F-13 attributions are two readings of one property. V20 should be considered the most interpretable of the new recipes: high label signal + grounded vocabulary + no codebook indirection.

*e) F-22 counterfactual under the extension.*: We do not re-run F-22 for every new recipe in this draft; the V20 counterfactual L1 ranking under LDA is competitive with V12 (median L1  $\geq 15$  on the labelled scenes where V20 has been profiled), suggesting that label-aware amplification produces topics that are also harder to perturb. A full F-22 ( $\times 7$  new recipes  $\times 6$  scenes) is left to a follow-up.

## VIII. RECOMMENDATION

For interpretability reporting, we recommend the following ordering:

- 1) **F-13 SHAP attribution.** Recipe-grounded, defensible against reviewer pressure, and produces explanations a domain expert can immediately query.
- 2) **F-22 counterfactual robustness.** Complements F-2/F-7 by ruling out trivial-overfit explanations.
- 3) **F-15 LLM-judge.** Only after vocabulary-size normalisation. Without normalisation it rewards trivial recipes.

The V13–V20 extension (§VII) further suggests a recipe-as-interpretability-tier classification:

- **Tier I (physically-grounded):** V1, V7, V14, V18, V20. Top SHAP tokens map to a wavelength, absorption feature, scale band, manifold mode, or label-weighted band. Use for any publishable interpretability claim.
- **Tier II (semantically-grounded):** V8, V10, V15. Top tokens map to endmember labels, group labels, or vegetation indices.
- **Tier III (learned-codebook-grounded):** V11, V12, V13, V17. Top tokens are codewords / GMM components / dictionary atoms. Interpretable only via a separate codebook-decoder visualisation.
- **Tier IV (abstract-coordinate):** V19. Topics point to UMAP coordinates with no spectral anchor. Use only as a sanity baseline.

## IX. DISCUSSION

**The three axes are complementary, not redundant.** F-13, F-22 and F-15 probe different facets of the same LDA fit. F-13 answers *which tokens explain a topic assignment*; F-22 answers *how robust that assignment is*; F-15 answers *whether a human-like judge would agree the document belongs to its topic*. The V20 recipe illustrates how they cohere: its MI-weighting concentrates both SHAP attribution mass (sharper F-13) and topic-word mass (lower  $N_{\text{eff}}$ , hence higher F-15 at fixed  $|\mathcal{V}|$ ), and the same concentration makes its topics the hardest to flip (F-22 = 26.3, the sweep maximum). This is consistent with P3’s finding that V20 leads F-2 / F-7 at  $Q = 32$ : a basis that concentrates discriminative mass is simultaneously more coherent, more robust, and more attributable.

**F-15 measures the metric, not the model.** The central methodological lesson is that an automated alignment proxy can fail for reasons that have nothing to do with topic quality. The anti-correlation between F-15 and F-2 on V3 / V12 is not evidence that those recipes produce incoherent topics — P3 ranks V12 *first* on F-2 and F-7 — but evidence that a top- $N$ -identity overlap rule is biased by token-mass dispersion ((18)). A genuine LLM oracle, given opaque token identifiers and asked for identity overlap, would inherit the same bias; only a judge equipped with *semantic* comparison of token meanings (e.g. the spectral region a token names) would escape it. This is a cautionary result for the growing practice of using LLMs to score topic models [3]: the prompt must expose token *semantics*, not token *strings*, or the score will track vocabulary engineering rather than topic content.

**Interpretability is a recipe property, not an LDA property.** The tier classification of §VIII shows that the same LDA backbone yields explanations ranging from directly physical (V1, V7, V14, V18, V20) to recursively opaque (V13, V17, V19). “LDA is interpretable” is therefore an incomplete claim; the defensible claim is “LDA on a physically-grounded wordification is interpretable.”

## X. LIMITATIONS

**LDA only.** Every result here is for the LDA backbone ((1)). The closed-form posterior that makes F-13 tractable, and the simplex-structured  $\phi_k$  on which  $N_{\text{eff}}$  is defined, do not transfer unchanged to ProdLDA’s product-of-experts decoder ((3)) or ETM’s embedding decoder ((4)); replicating these axes under P4’s backbones is open work.

**Self-judge stand-in for the LLM oracle.** F-15 was produced by a deterministic encoding of the alignment rule (Claude Opus 4.7, 1M context) rather than live per-call API judging. This makes the run reproducible and free, but it cannot exhibit the semantic-comparison behaviour that a true oracle might; the API builder (`build_v_sweep_f15_llm_alignment.py`) is provided for users who wish to substitute live judging. Our claims about F-15 concern the *metric definition* ((17)) and hold regardless of which judge instantiates it.

$N_{\text{eff}}$  **is computed on  $\phi_k$ .** We measure dispersion on the topic-word distributions, the quantity the topic-side of the F-15 rule reads; the canonical metric ((18)) is stated on a generic

mass vector  $p_d$  and applies equally to per-document counts. The two give the same qualitative ordering for the decisive V3 / V12 / V20 comparison but differ in absolute scale; we report the  $\phi_k$  version because it is what F-15 actually depends on.

**Partial F-22 under the extension.** The full F-22 table (Table I) is complete for the  $Q = 8$  sweep, but the  $\times Q \times$  new-recipe cross-product (§VII) is profiled only spot-wise; a full re-run is left to a follow-up.

**Six labelled scenes.** The HIDSAG mineralogical subsets used in P1/P3 are not included here because their per-pixel label structure differs; the dispersion result should be re-checked on them before being claimed as scene-independent.

## XI. CONCLUSION

We instantiated three post-hoc interpretability axes — F-13 SHAP attribution, F-22 counterfactual  $L_1$ , and F-15 LLM-judge alignment — on the LDA wordification sweep of P3, and reported three results. First, F-13 SHAP gives recipe-grounded explanations whose physical meaning is fixed by the recipe and cannot silently shift across scenes; this is the axis we recommend as the primary interpretability artefact. Second, F-22 separates the recipes into robust (V20 = 26.3, V12 = 24.5, V3 = 23.5) and fragile (V9 = 1.0) topic-boundary regimes, complementing P3’s coherence ranking by ruling out a trivial-overfit account of V20’s lead. Third, and most important methodologically, F-15 is governed by *token-mass dispersion*  $N_{\text{eff}} = \exp(H)$  ((18)), *not* nominal vocabulary cardinality: among the three recipes whose alphabets are the same size (mean  $|\mathcal{V}| \approx 1376$ ; V3/V20 identical (band, bin), V12 same-sized GMM), F-15 rises (0.12  $\rightarrow$  0.16  $\rightarrow$  0.64) exactly as topic dispersion falls (431  $\rightarrow$  400  $\rightarrow$  309), while a cardinality account predicts no difference. We therefore recommend reporting F-15 only after dispersion normalisation, and we caution that LLM-as-judge topic evaluation inherits the same artefact unless the judge is given token *semantics* rather than token strings. Extending these axes across the P4 backbones, and re-checking the dispersion result on the HIDSAG subsets, are the natural next steps.

## REFERENCES

- [1] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017, pp. 4765–4774, introduces SHAP (SHapley Additive exPlanations) and the KernelExplainer; the additive feature-attribution framework used for the F-13 per-pixel per-topic attribution maps in this manuscript. [Online]. Available: <https://papers.nips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>
- [2] S. Wachter, B. Mittelstadt, and C. Russell, “Counterfactual explanations without opening the black box: Automated decisions and the GDPR,” *Harvard Journal of Law & Technology*, vol. 31, no. 2, pp. 841–887, 2017, foundational definition of counterfactual explanations as the minimal feature perturbation that flips a model’s decision; the conceptual basis for the F-22 minimum- $L_1$  topic-flip metric. [Online]. Available: <https://jolt.law.harvard.edu/assets/articlePDFs/v31/Counterfactual-Explanations-without-Opening-the-Black-Box-Sandra-Wachter-et-al.pdf>
- [3] D. Stambach, V. Zouhar, A. Hoyle, M. Sachan, and E. Ash, “Revisiting automated topic model evaluation with large language models,” in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023, pp. 9348–9357, shows that large language models can serve as automated judges of topic quality (word-intrusion and coherence-rating tasks) that correlate with human judgement; the LLM-as-judge basis for the F-15

- alignment axis and the B-12 tea-leaves probe. [Online]. Available: <https://aclanthology.org/2023.emnlp-main.581/>
- [4] M. Wahabzada, A.-K. Mahlein, C. Bauckhage, U. Steiner, E.-C. Oerke, and K. Kersting, "Plant phenotyping using probabilistic topic models: Uncovering the hyperspectral language of plants," *Scientific Reports*, vol. 5, p. 22482, 2015, earliest LDA-style probabilistic topic model on hyperspectral plant-phenotyping data; clarifies the wiki References §4 "Scientific Reports / PLOS ONE lineage" framing.
  - [5] S. Zou and A. Zare, "A continuous-space partial-membership latent Dirichlet allocation approach for subpixel hyperspectral imagery," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 55, no. 10, pp. 5840–5853, 2017.
  - [6] L. Liu, C. Wang, B. Du, and H. Lai, "A latent Dirichlet allocation-based local spectral-spatial hyperspectral image classification method," *IEEE Geoscience and Remote Sensing Letters*, vol. 16, no. 6, pp. 935–939, 2019.
  - [7] F. A. Santibañez-Leal, Á. F. Egaña, A. Ehrenfeld, F. Navarro, and B. Townley, "Geometallurgical estimation of mineral samples from hyperspectral images and statistical topic modelling," in *Proceedings of the 18th International Conference on Mineral Processing and Geometallurgy (Procemin Geomet 2022)*. Santiago, Chile: Gecamin, 2022, internally referred to as the "A39" paper. Introduces the LDA-V1/V2/V3 spectral-document recipes and the topic-routed hierarchical regressor that reduced copper-recovery MAE from  $\sim 4.57$  (per-spectrum baseline) to  $\sim 0.42$  on private mineral data. [Online]. Available: <https://www.researchgate.net/publication/369708272>
  - [8] D. M. Blei, A. Y. Ng, and M. I. Jordan, "Latent Dirichlet allocation," *Journal of Machine Learning Research*, vol. 3, pp. 993–1022, 2003. [Online]. Available: <https://jmlr.org/papers/v3/blei03a.html>
  - [9] M. D. Hoffman, D. M. Blei, and F. R. Bach, "Online learning for latent Dirichlet allocation," in *Advances in Neural Information Processing Systems 23 (NIPS 2010)*, 2010, pp. 856–864. [Online]. Available: <https://papers.nips.cc/paper/3902-online-learning-for-latent-dirichlet-allocation>
  - [10] D. Newman, J. H. Lau, K. Grieser, and T. Baldwin, "Automatic evaluation of topic coherence," in *Human Language Technologies: The 2010 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, 2010, pp. 100–108, foundation for the human-judgement-correlated PMI / NPMI coherence definitions that Röder et al. 2015 generalises. [Online]. Available: <https://aclanthology.org/N10-1012/>
  - [11] D. Mimno, H. M. Wallach, E. Talley, M. Leenders, and A. McCallum, "Optimizing semantic coherence in topic models," in *Proceedings of the 2011 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2011, pp. 262–272, source of the U-Mass document-co-occurrence coherence statistic used in the  $c_v$  / NPMI / U-Mass battery. [Online]. Available: <https://aclanthology.org/D11-1024/>
  - [12] M. Röder, A. Both, and A. Hinneburg, "Exploring the space of topic coherence measures," in *Proceedings of the Eighth ACM International Conference on Web Search and Data Mining (WSDM)*, 2015, pp. 399–408.
  - [13] D. Greene, D. O'Callaghan, and P. Cunningham, "How many topics? Stability analysis for topic models," in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases (ECML-PKDD)*, ser. Lecture Notes in Computer Science, vol. 8724. Springer, 2014, pp. 498–513, multi-seed Hungarian-matching topic-stability protocol used in the seed-stability validation step of this project. [Online]. Available: <https://arxiv.org/abs/1404.4606>
  - [14] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. [Online]. Available: <https://jmlr.org/papers/v12/pedregosa11a.html>
  - [15] M. F. Baumgardner, L. L. Biehl, and D. A. Landgrebe, "220 band AVIRIS hyperspectral image data set: June 12, 1992 Indian Pine test site 3," Purdue University Research Repository, 2015, indian Pines AVIRIS hyperspectral dataset, calibrated 220-band reflectance, ground-truth labels for 16 land-cover classes.
  - [16] NASA/JPL Airborne Visible/Infrared Imaging Spectrometer Project, "AVIRIS hyperspectral imagery — Salinas valley, california," [https://www.ehu.eu/ccwintco/index.php/Hyperspectral\\_Remote\\_Sensing\\_Scenes#Salinas\\_scene](https://www.ehu.eu/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Salinas_scene), 1998, 224-band hyperspectral cube, 3.7 m spatial resolution. Calibrated and corrected version maintained by University of Pavia GIC. Salinas-A is a  $86 \times 83$  sub-scene with 6 classes. [Online]. Available: [https://www.ehu.eu/ccwintco/index.php/Hyperspectral\\_Remote\\_Sensing\\_Scenes#Salinas\\_scene](https://www.ehu.eu/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Salinas_scene)
  - [17] German Aerospace Center (DLR) ROSIS sensor, "ROSIS Pavia University hyperspectral dataset," [https://www.ehu.eu/ccwintco/index.php/Hyperspectral\\_Remote\\_Sensing\\_Scenes#Pavia\\_Centre\\_and\\_University](https://www.ehu.eu/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_Centre_and_University), 2003, reflective Optics System Imaging Spectrometer airborne acquisition over Pavia, Italy. 103 bands after bad-band removal, 9 classes,  $610 \times 340$  pixels at 1.3 m resolution. [Online]. Available: [https://www.ehu.eu/ccwintco/index.php/Hyperspectral\\_Remote\\_Sensing\\_Scenes#Pavia\\_University\\_scene](https://www.ehu.eu/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_University_scene)
  - [18] NASA/JPL Airborne Visible/Infrared Imaging Spectrometer Project, "AVIRIS Kennedy Space Center hyperspectral imagery," [https://www.ehu.eu/ccwintco/index.php/Hyperspectral\\_Remote\\_Sensing\\_Scenes#Kennedy\\_Space\\_Center\\_28KSC.29](https://www.ehu.eu/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Kennedy_Space_Center_28KSC.29), 1996, Kennedy Space Center, Florida, 18 m resolution, 224 bands, 13 land-cover classes after labelling. [Online]. Available: [https://www.ehu.eu/ccwintco/index.php/Hyperspectral\\_Remote\\_Sensing\\_Scenes#Kennedy\\_Space\\_Center\\_28KSC.29](https://www.ehu.eu/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Kennedy_Space_Center_28KSC.29)
  - [19] NASA EO-1 Hyperion Mission, "EO-1 Hyperion Okavango Botswana hyperspectral imagery," [https://www.ehu.eu/ccwintco/index.php/Hyperspectral\\_Remote\\_Sensing\\_Scenes#Botswana](https://www.ehu.eu/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Botswana), 2001, hyperion satellite-borne hyperspectral sensor, Okavango Delta, Botswana. 145 bands after preprocessing, 14 land-cover classes. [Online]. Available: [https://www.ehu.eu/ccwintco/index.php/Hyperspectral\\_Remote\\_Sensing\\_Scenes#Botswana](https://www.ehu.eu/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Botswana)